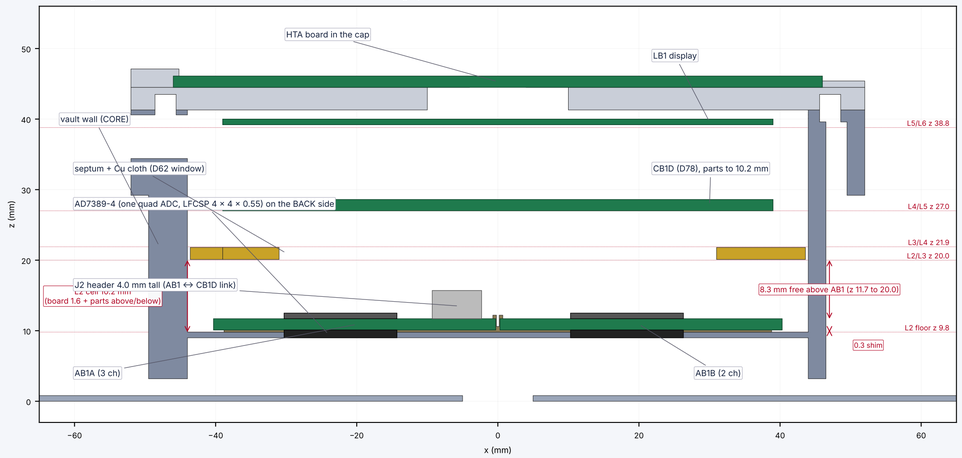
PHI-FOLD 233 receive stage cut-away: AB1A / AB1B with the AD7389-4 and the +3V3D island (MODELED, tool-checked, nothing built)

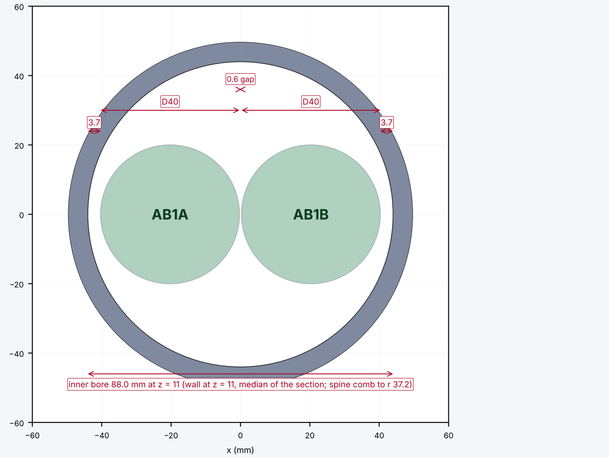
1 Vault half-section (y = 0), cut faces from the current meshes

heel (178 deg) to the left; boards drawn as slabs at the DESIGN_STATE section 3 z-ranges; D40 AB1A/AB1B at x = -/+ 20.3



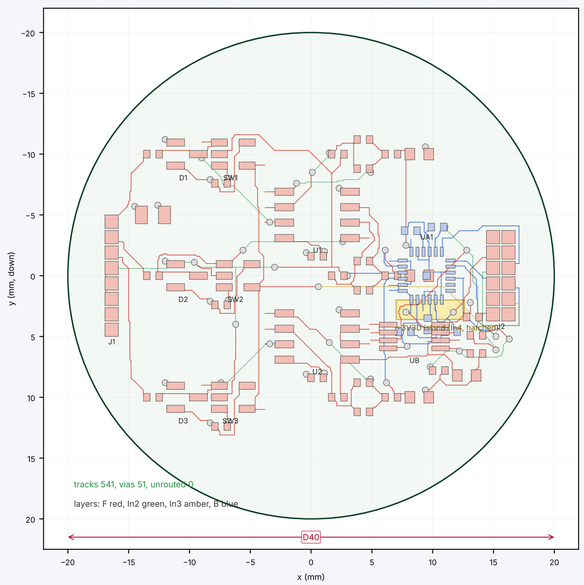
2 Plan section at z = 11 (AB1 level): what the two D40 discs must fit inside

ring = the core wall cut at z = 11; discs AB1A / AB1B at x = -/+ 20.3; the shared strip between them carries J2 and the BLANK / SPI links



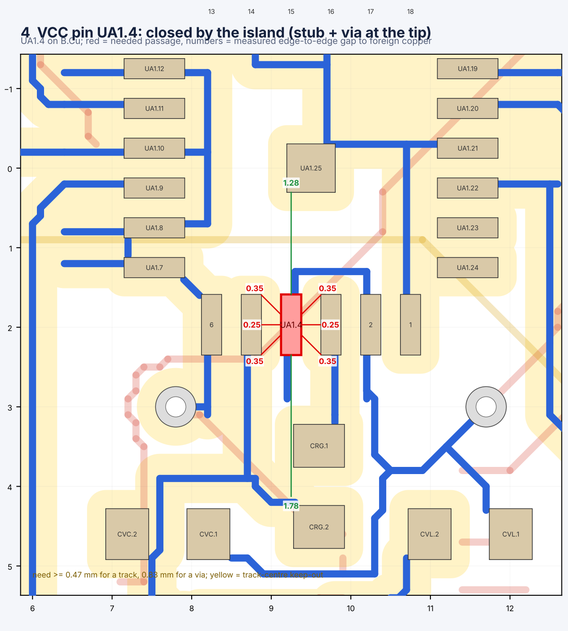
3 AB1A routed layout (AD7389-4 build): 0 open nets; the +3V3D island on In4 is the yellow box

board frame, y down; tracks coloured by layer; planes In1 = GND and In4 = +5VA (hatched) are not drawn; island x 7.0..12.6, y 2.0..3.6 mm



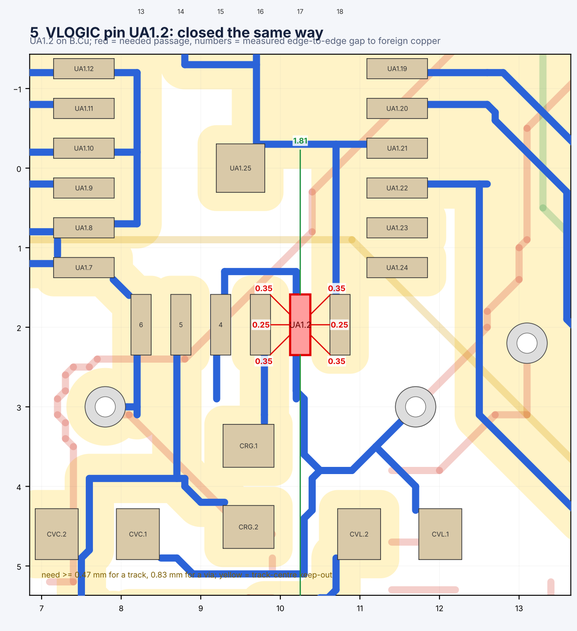
4 VCC pin UA1.4: closed by the island (stub + via at the tip)

UA1.4 on B.Cu; red = needed passage, numbers = measured edge-to-edge gap to foreign copper



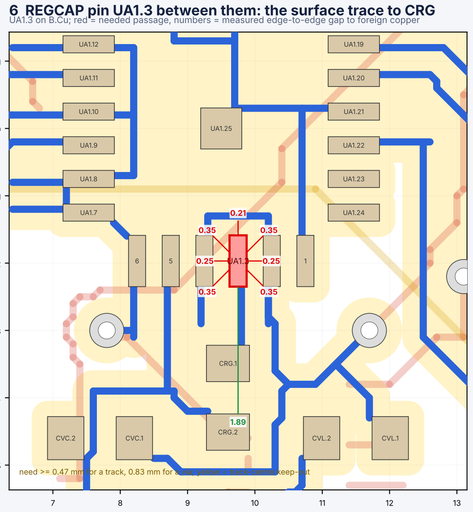
5 VLOGIC pin UA1.2: closed the same way

UA1.2 on B.Cu; red = needed passage, numbers = measured edge-to-edge gap to foreign copper



6 REGCAP pin UA1.3 between them: the surface trace to CRG

UA1.3 on B.Cu; red = needed passage, numbers = measured edge-to-edge gap to foreign copper



7 Receive-board status and every item still open

item	state / measured number	reason / next step
AB1A routing	0 unrouted nets, 541 tracks, 51 vias (two identical runs); KiCad DRC 0 errors / 0 unconnected	was 3 open nets with 3 x ADS8860; closed by one AD7389-4 and the +3V3D island
AD7389-4 pin row	pad 0.25 × 0.76 mm, pitch 0.50 mm: pad gap 0.25	a track needs 0.47 mm, a via 0.83 mm: pins 2 and 4 leave by a stub and a via into the island
+3V3D island (In4)	x 7.0..12.6, y 2.0..3.6 mm, 8.96 mm2, hatched; Gerber read-back: island vias connected, 0.16 mm to the +5VA pour	3.025 mm from the OPA2211 pads; no closed copper loop around the axis
Fab floor (AB1 override)	clearance 0.16, track 0.15, via 0.508 / drill 0.254	TTM T-HDI Standard, AdvancedPCB A-TOL; U-FAB-DRC: some facilities list 0.216 ring
Router	greedy single pass A*, 0.1 mm grid, no rip-up	enough for this build; Options 2 (rip-up) and 3 (J2 move) were not needed
J2 pad block	2 × 6, 1.27 mm pitch, pads 1.10 mm (ring 0.225, gap 0.17)	J2.3 (+3V3D) reaches the island by one trace and a via; SDO3 unused (2-wire readout)
AB1B	0 unrouted, KiCad DRC 0 errors / 0 unconnected, independent DRC 0 fails	same chip and island; MONO input pad 3.085 mm from the island
Datasheet pin maps	AD7389-4 read from the supplied datasheet; SW1G66, OPA2365, BAV199 marked VERIFY	exposed pad size of the LFCSP is a PLACEHOLDER (drawing shows none, text says ground it)
Mechanics	CORE cylinder + 14 other STLs non-manifold; U8_SEPTUM_extended oversize	vault +8 mm (option 2) and rim disc not implemented; no eddy / field model for an added board
Unverified	SDI tied low, +3V3D rail budget (about 25 mA more), island hatch in a CAM viewer, fabricator DRC, bench	U-DS-TXDRIVE, OKI, PACK, BAV199, HVTERM, U-FAB-DRC, U-CAM-VIEW, U-BENCH